

Mohamed Youssef

(917) 909-9858 | mohamed.youssef@nyu.edu | [linkedin.com/in/mohamed-youssef16](https://www.linkedin.com/in/mohamed-youssef16) | mohamedyoussef-glass.netlify.app | Brooklyn, NY

EDUCATION

New York University Tandon School of Engineering Expected 2028
Electrical and Computer Engineering, GPA: 3.67/4.0 Brooklyn, NY
Coursework: Signals & Systems, Circuits, Digital Logic, Engineering and Design, Object-Oriented Programming, Data Structures and Algorithms, Computer Architecture, Databases
Honors: Tandon Honors Program: Global Leaders and Scholars in STEM, a cohort of 25 students selected for academic excellence, leadership, and research potential
Study Abroad: NYU Abu Dhabi (UAE); Hanyang University (South Korea)
Clubs & Programs: HEOP, TRIO Scholars, SHPE, MSA, NAS

TECHNICAL SKILLS

Programming: Python, C++, HTML, CSS
Simulation: Ansys HFSS, Keysight Advanced Design System (ADS), KiCad, Altium, LTspice, MatLab, Verilog
Prototyping / Hardware: Arduino, Raspberry Pi, Fusion 360, LPKF PCB Milling, 3D Printing, Laser Cutting

EXPERIENCE

Teaching Assistant Jan 2025 – Present
NYU MakerSpace Brooklyn, NY

- Serve as LPKF PCB specialist, training students in Gerber preparation, substrate setup, and machine calibration
- Maintain lab safety and accessibility while assisting 15+ students daily
- Collaborate with manager and 3-member training team to develop 5+ machine operation safety guides and FAQs
- Design and lead Raspberry Pi workshop teaching 10+ students introductory programming and hardware skills

PCB Engineer Sep 2026 – Present
Robotics Design Team Brooklyn, NY

- Coordinate with multidisciplinary engineering teams to define PCB design specifications and integrate them into hardware development.
- Designed a competition-compliant PCB in accordance with all NASA Lunabotics Competition design standards and technical requirements

Electrical Engineering Intern Jul 2026 – Aug 2026
Dot Inc. via Hanyang University Seoul, South Korea

- Conducted blindfolded field testing at museums to evaluate user experience and inform product decisions
- Built a real-time image-to-tactile pipeline linking smart glasses to a Dot Pad display across 4 processing methods
- Developed an iOS capture app for real-time museum artwork rendering
- Field-tested hardware across 2 museums and 15+ exhibits to define accessibility requirements and benchmark accuracy
- Presented prototype demonstrations and testing results to engineers, incorporating feedback into future iterations

Lifeguard / Swim Instructor Jun 2022 – Sep 2025
YMCA of Greater New York Queens, NY

- Led weekly swim instruction for 50+ students of diverse backgrounds by tailoring lessons to individual needs
- Monitored pool safety, resulting in 0 serious injuries while on duty

PROJECTS

Wireless 6G | *NYU Wireless, Ansys HFSS, Keysight ADS, KiCad, Altium LPKF PCB Milling*

- Design and simulate 4 high-frequency microstrip transmission lines using Ansys HFSS and Keysight ADS
- Compare simulated and measured data to refine material parameters and improve RF modeling accuracy by 10%
- Model and troubleshoot electromagnetic behavior using S-parameter analysis
- Manufacture all 4 simulation results as physical prototypes via LPKF PCB milling
- Investigate fabrication tolerances including solder quality and connector torque to improve measurement repeatability

Motion-Detecting Scarecrow | *Arduino C++, 3D Printing*

- Collaborated with a 4-member team to develop a functional prototype and coordinate task assignments
- Built an Arduino C++ scarecrow prototype on a breadboard with 5+ components
- Managed a \$100 budget using 3D-printed and off-the-shelf parts

Portfolio Website | *HTML, CSS*

- Developed a responsive portfolio website using HTML, and CSS to showcase projects and experiences
- Created a reusable CSS design system with variables for consistent styling across the site
- Implemented interactive features including tabbed content, scroll-spy navigation, and a mobile menu